

SAN SEBASTIAN IGUELDO, Spain

WMO: 080270

Lat: 43.3064N

Lon: 2.0411W

Elev: 259

StdP: 98.25

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.1	(c) 1.3	(d) -4.9	(e) 2.6	(f) 4.3	(g) -3.2	(h) 3.0	(i) 4.8	(j) 17.3	(k) 11.3	(l) 15.4	(m) 11.0	(n) 2.6	(o) 160	(p) 0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.7	(c) 28.4	(d) 20.4	(e) 26.0	(f) 19.3	(g) 24.0	(h) 18.7	(i) 21.9	(j) 25.8	(k) 20.9	(l) 23.7	(m) 20.1	(n) 22.4	(o) 5.0	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	15.9	23.4	19.9	15.1	22.1	19.2	14.5	21.2	65.2	25.9	61.5	23.8	58.7	22.4	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a) 14.2	(b) 12.0	(c) 10.3	(d) DB	(e) -1.8	(f) 34.1	(g) 1.7	(h) 2.4	(i) -3.0	(j) 35.8	(k) -4.0	(l) 37.2	(m) -5.0	(n) 38.5	(o) -6.3	(p) 40.2	
(a) 14.2	(b) 12.0	(c) 10.3	(d) WB	(e) -2.9	(f) 24.4	(g) 1.9	(h) 0.8	(i) -4.3	(j) 25.0	(k) -5.5	(l) 25.5	(m) -6.5	(n) 25.9	(o) -7.9	(p) 26.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	8.5	8.7	10.3	11.8	14.3	17.2	18.9	19.5	17.7	15.8	11.4	9.3
	DBStd	5.07	3.45	3.79	3.58	3.12	3.09	2.97	2.33	2.35	2.70	3.55	3.50	3.54
	HDD10.0	264	69	62	38	13	2	0	0	0	0	2	23	54
	HDD18.3	1863	306	271	249	198	131	55	18	11	43	95	207	279
	CDD10.0	1599	22	26	48	67	136	217	276	294	232	182	67	33
	CDD18.3	156	0	0	1	2	7	22	36	47	25	16	1	0
	CDH23.3	605	0	0	0	12	44	144	118	169	91	26	1	0
	CDH26.7	165	0	0	0	1	10	50	41	49	14	1	0	0
(14) Wind	WSAvg	4.1	5.0	4.9	4.7	4.3	3.9	3.3	3.1	3.0	3.3	4.3	5.0	4.9
(15) Precipitation	PrecAvg	1598	157	128	118	130	123	101	92	116	105	136	212	145
	PrecMax	2146	370	323	238	266	265	257	188	280	222	328	405	253
	PrecMin	1249	6	18	33	49	62	24	38	55	37	50	45	16
	PrecStd	228	94	84	56	62	60	53	46	65	42	75	86	70
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	19.6	21.8	25.0	27.9	31.8	31.5	30.7	28.2	25.7	20.7	18.3
		MCWB	11.1	10.5	12.5	16.0	17.9	21.5	21.4	22.2	20.2	18.3	14.4	12.2
	2%	DB	15.7	17.2	19.8	21.3	23.8	27.3	26.3	27.7	25.9	23.7	19.2	16.7
		MCWB	11.1	10.1	12.0	14.0	16.4	19.4	20.8	21.0	18.8	17.3	13.7	11.5
	5%	DB	14.5	15.3	17.8	18.5	21.4	23.8	23.5	24.7	23.7	22.4	17.8	15.2
		MCWB	10.3	9.7	11.1	12.6	15.2	18.4	20.2	20.2	18.3	16.4	13.2	10.8
	10%	DB	13.1	13.6	15.8	16.5	19.2	21.1	21.8	22.6	21.6	21.0	16.4	14.0
		MCWB	9.6	9.1	10.3	11.8	14.6	17.8	19.5	19.8	18.0	15.8	12.3	10.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	12.7	14.0	16.5	19.2	22.6	23.1	23.5	21.7	19.3	16.0	13.6
		MCDB	15.8	16.4	19.7	23.9	25.9	29.8	27.3	27.5	25.6	23.1	18.2	15.9
	2%	WB	11.9	11.6	12.7	14.6	17.4	20.2	21.5	22.1	20.4	18.1	14.7	12.5
		MCDB	14.2	14.6	17.9	19.2	21.5	24.3	24.9	25.6	23.4	21.6	17.6	14.9
	5%	WB	11.1	10.8	11.8	13.5	16.2	19.1	20.5	21.0	19.6	17.3	13.8	11.6
		MCDB	13.1	13.4	15.8	16.8	19.6	22.5	22.9	23.5	22.0	20.6	16.5	14.2
	10%	WB	10.2	10.0	11.0	12.6	15.3	18.2	19.7	20.2	18.7	16.6	13.0	10.9
		MCDB	12.2	12.3	14.2	15.1	18.0	20.6	21.5	22.1	20.8	19.6	15.5	13.4
(35) Mean Daily Temperature Range	5% DB	MDBR	3.9	4.6	4.9	5.0	5.1	4.8	4.3	4.7	4.9	4.8	4.0	4.0
		MCDBR	5.3	7.0	8.3	9.1	10.2	10.7	8.4	9.3	8.5	6.9	5.5	5.4
		MCWBR	3.3	3.8	4.3	4.6	5.0	5.3	4.5	4.8	4.5	3.7	3.1	3.4
	5% WB	MCDBR	4.5	5.8	6.5	7.5	8.1	8.4	6.8	7.1	6.6	6.0	4.8	4.7
		MCWBR	3.3	3.8	3.9	4.3	4.6	4.7	3.9	4.1	4.0	3.6	3.3	3.6
(40) Clear-Sky Solar Irradiance	taub		0.320	0.335	0.367	0.390	0.400	0.411	0.402	0.401	0.380	0.358	0.335	0.316
	taud		2.514	2.462	2.380	2.325	2.334	2.335	2.363	2.375	2.424	2.488	2.503	2.521
	Ebn at Noon		816	862	871	876	874	862	865	855	847	823	788	785
	Edh at Noon		71	87	107	121	124	124	119	114	101	84	71	64
(44) All-Sky Solar Radiation	RadAvg		1.42	2.26	3.49	4.55	5.42	6.00	6.02	5.32	4.28	2.78	1.66	1.32
	RadStd		0.17	0.30	0.43	0.47	0.42	0.38	0.32	0.33	0.22	0.23	0.20	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page